SHREYAS KARNAD

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PROFESSIONAL SUMMARY

Health Informatics graduate of the University of Michigan (MHI, 3.936 GPA) and Biomedical Engineer with expertise in machine learning, generative AI pipelines, and healthcare data analytics. Built multi-agent LLM systems using CrewAI and DeepSeek, developed predictive models for clinical risk stratification, and designed FHIR-interoperable data workflows. Bridges technical depth in Python, R, and SQL with applied knowledge of clinical decision support, EHR systems, HIPAA compliance, and health data governance.

EDUCATION

University of Michigan – School of Information, Ann Arbor, MI Aug 2024 – May 2026
Master of Health Informatics (Machine Learning, Data Analytics) - GPA: 3.936
Focus: health data analytics, clinical decision support, FHIR interoperability, AI/ML in healthcare, data governance

Manipal Institute of Technology, MAHE, Manipal, India Oct 2021 – Aug 2023
M.Tech, Biomedical Engineering (Biomedical Instrumentation) - GPA: 3.87

MIT School of Bioengineering Sciences & Research, MIT ADT, Pune, India Aug 2017 – Aug 2021
B.Tech, Bioengineering (Biomedical Engineering elective) - GPA: 3.93

EXPERIENCE

University of Michigan – School of Public Health, Ann Arbor, MI Jan 2026 – May 2026
Grader, Public Health Informatics (Part-time)

- Evaluated 98+ graduate-level assignments per cycle using structured rubrics, maintaining consistent scoring standards and reducing instructor review time through detailed written feedback on analytical approach and data interpretation.
- Collaborated directly with Prof. Rahul Ladhania on grading rubric design, dispute resolution, and quality assurance across multiple assignment cycles; identified and corrected systematic grading inconsistencies before grade release.

University of Michigan – School of Information, Ann Arbor, MI Sep 2025 – Dec 2025
Instructional Aid, Health Informatics / Database Management (Part-time)

- Supported a 150+ student health informatics course by grading analytical examinations, proctoring assessments, and communicating complex technical concepts clearly to students with varying programming backgrounds.
- Collaborated directly with Prof. Bobby Madamanchi on exam paper design, dispute resolution, and quality assurance across 3 exam cycles; identified and corrected systematic grading inconsistencies before grade release.

Biocomplexity Institute, Indiana University, Bloomington, IN (Remote) Jun 2025 – Aug 2025
Data Science Intern

- Designed schema-based JSON pipelines to automatically generate syntactically valid CompuCell3D biological simulation code from research paper abstracts; documented zero-shot LLM failure modes (version mismatches, hallucinated APIs, incorrect biological parameters) to guide iterative pipeline refinement.
- Translated natural-language biological modeling questions into structured computational parameters (adhesion, chemotaxis, diffusion) via an NL → ontology → code framework, enabling non-programmer researchers to generate simulation-ready outputs.
- Re-engineered a 9-agent CrewAI system (Scientist, Team Leader, Coder, Ontology Expert, Biological Modeling Expert, Version Checker, Coverage Verification, Packager, Interpreter) to replace OpenAI with DeepSeek Replicate via a custom Python wrapper in VS Code, resolving API environment conflicts and reducing inference costs while maintaining code generation accuracy.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India (Remote) Aug 2023 – Aug 2024
Associate Biomedical Engineer

- Evaluated deep learning classification models (ResNet50, GoogLeNet, DenseNet) in MATLAB Deep Learning Toolbox to classify harmonic signature data from magnetic nanoparticle-based intraoperative sentinel lymph node detection; achieved 96–100% accuracy distinguishing tumor-present from tumor-absent nodes in early-stage testing, with bacterial species identification validated as a secondary application.
- Integrated MATLAB with LabVIEW to interface with MafPro magnetic field probe hardware; deployed ML inference pipelines to AWS S3-connected cloud servers, enabling remote clinics without high-end NPU hardware to offload classification tasks to cloud infrastructure and receive results in real time.
- Designed a ruggedized field enclosure in Fusion 360 housing a Surface Pro with NPU, battery pack, and CAT8 magnetic detector port; implemented a locked-down Windows 11 environment restricting end-user access to MafPro software only, producing a deployment-ready clinical device prototype.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India Aug 2022 – Jul 2023
Graduate Research Intern

- Conducted structured primary research with 20+ surgical oncologists to identify unmet clinical workflow needs; synthesized findings into formal requirements that directly shaped the product development roadmap.

- Built and validated a magnetic probe prototype for intraoperative sentinel lymph node detection; analyzed experimental sensor data using time-series methods and contributed findings to M.Tech thesis and internal technical reporting.

TATA Elxsi, St. Paul, MN (Remote) Jan 2021 – Jun 2021

Project Intern

- Analyzed investment data across 50+ digital health companies totaling over $12B; built weighted scoring models to identify high-growth sectors including AI/ML, cloud computing, and IoT, delivering sector rankings to support strategic investment decisions.
- Mapped Software as a Medical Device (SaMD) regulatory frameworks across FDA and IMDRF pathways, producing a risk classification reference guide used to inform product development planning.

PROJECTS

EHR AI Hallucination Mitigation Strategy Mar 2026 – Apr 2026

- Identified that AI-generated hallucinations in EHR transcription tools create undetected clinical documentation errors; proposed a Zero-Trust RAG framework with retrieval-based validation checkpoints, reducing reliance on unverified LLM outputs and improving documentation accuracy and HIPAA compliance.

Ctrl Alt Drink - Mobile Health Intervention for Student Alcohol Moderation Jan 2026 – Apr 2026

- Designed a behavioral health app targeting student alcohol moderation; incorporated real-time behavioral tracking, anonymous peer support, and micro-goal frameworks grounded in evidence-based behavior change principles to address a gap in accessible, stigma-free campus health tools.

Structural Health Disparities and Sepsis Risk Analysis Sept 2025 – Dec 2025

- Analyzed a large administrative California county-level healthcare dataset to determine structural drivers of sepsis risk; built a Random Forest classification model achieving 80% accuracy and multivariate regression identifying insurance coverage as the primary predictor of postoperative outcomes.
- Authored full statistical methods and results sections for academic submission, including data quality validation, feature engineering, and county-level health disparity visualizations.

Python-Based Diabetes Risk Assessment App + FHIR Output Oct 2024 – Nov 2024

- Built an end-to-end ML pipeline for diabetes risk prediction; generated structured FHIR-formatted outputs to enable interoperable data sharing across health systems, demonstrating real-world applicability of clinical decision support tools.

Diabetes Risk Analysis - SI 544, University of Michigan Sep 2024 – Dec 2024

- Applied hypothesis testing (two-sample t-tests) and multivariate linear regression in R to a 768-patient Pima Indians diabetes dataset; identified blood glucose ($p < 2.2e\text{-}16$) and BMI as statistically significant primary predictors while surfacing zero-value missingness as a systematic data quality issue affecting model reliability.

Breast Cancer Tumor Network Analysis - SI 507, University of Michigan Feb 2025 – Apr 2025

- Compared K-Means and Spectral Clustering algorithms on the Wisconsin Diagnostic breast cancer dataset (32 features, 569 patients); Spectral Clustering outperformed on accuracy and runtime, while network centrality analysis identified `concave_points_mean` as the most diagnostically influential tumor feature across degree, closeness, and betweenness metrics.

IVF Success Prediction - Binary Classification, Manipal Institute of Technology April 2022 – June 2022

- Applied Logistic Regression, K-Nearest Neighbour, and Naïve Bayes classifiers in Python to a clinical IVF dataset (55 patients, 14 metabolites); KNN achieved best accuracy of 83% on optimized metabolite combinations; identified data volume as primary constraint on model reliability.

PPI2 Web Server - Protein-Protein Interaction Database Jan 2019 – Dec 2020

- Developed an HTML-based research data system interfacing with a relational biological database; integrated interactive 3D protein interaction visualizations to support researcher queries and data exploration.

PAPERS

Current Trends of AI and Applications in Digital Pathology: A Comprehensive Review Dec 2023

- *International Journal of Advanced and Applied Sciences (IJAAS)* - peer-reviewed publication

TECHNICAL SKILLS

Programming: Python (pandas, scikit-learn, matplotlib, seaborn), R (tidyverse, ggplot2, caret), SQL, MATLAB, LabVIEW | **GenAI/LLM:** CrewAI, OpenAI APIs, DeepSeek Replicate, prompt engineering, multi-agent workflows, RAG, ontology extraction | **ML/Stats:** predictive modeling, classification, regression, clustering, hypothesis testing, biostatistics, time-series analysis, harmonic signal analysis | **Deep Learning:** ResNet50, GoogLeNet, DenseNet, MATLAB Deep Learning Toolbox, image classification, biological signal classification | **Healthcare/Interop:** FHIR (file generation), HL7, HIE architecture, EHR systems, clinical decision support, HIPAA, 42 CFR | **Data:** MySQL, Microsoft Access, Excel; data pipeline design, EDA, feature engineering, data quality assurance | **Cloud/Tools:** AWS (S3, server-based inference), Git, VS Code, Fusion 360; technical documentation, reproducible workflows | **Languages:** English (fluent), Hindi/Hindustani (fluent), Marathi (native)